

Exceptional-Variation Abel Return: Literal Weights, Almost-Prefixes, and the Atomic All-Prefix Barrier

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Abstract

TPC-159 controls cumulative determinant-two Möbius sums at every endpoint outside a low range and an explicit dyadic exceptional shadow. We combine that result with exact Abel summation. A literal weight is charged by its variation on good and bad endpoints: good variation receives the logarithmic correlation saving, while bad variation receives only the trivial bound. This gives a sharp conditional route for physical weights. For a prefix cutoff the Abel derivative is one atom, so uniform all-prefix control requires the actual endpoint registry to avoid the shadow or a new pointwise theorem. Exceptional-set density alone cannot provide this.

Keywords. Abel summation; Möbius correlation; literal weight; exceptional variation; atomic prefix.

Claim boundary. The Abel inequality is proved on the actual periodic two-Möbius core. A decaying physical-weight estimate remains conditional on a source-locked literal weight and quantitative variation bounds. The all-prefix obstruction stops only this exceptional-shadow/Abel route. No generic phase, four-sign H3 return, positive fixed- X power, $1/400$, prime-pair lower bound or twin-prime theorem is proved.

1 Ordered fiber and the bad set

Retain

$$q = as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az),$$

and a bounded period- R function ρ . Let

$$0 < t_1 < t_2 < \cdots < t_m \leq T$$

be the points $t(z_i)$, and put

$$\sigma_i = c_{z_i}\rho(z_i), \quad A_i = \sum_{j=1}^i \sigma_j.$$

TPC-159 fixes

$$J = \lceil A \log_2 \log X \rceil, \quad \mathcal{S}_{X,J} = \bigcup_{j=1}^J 2^j \mathcal{E}_X^*$$

and proves a prefix estimate outside that shadow [4]. Define

$$\mathcal{B}_{X,J} = [1, 2^J \sqrt{X}) \cup \mathcal{S}_{X,J}. \tag{1}$$

2 Exact Abel summation

For arbitrary literal weights $w_1, \dots, w_m \in \mathbb{C}$, set

$$d_i = w_i - w_{i+1} \quad (1 \leq i < m), \quad d_m = w_m. \quad (2)$$

Lemma 2.1 (Finite Abel identity).

$$\sum_{i=1}^m \sigma_i w_i = \sum_{i=1}^m A_i d_i. \quad (3)$$

Proof. The coefficient of σ_j on the right is

$$\sum_{i=j}^{m-1} (w_i - w_{i+1}) + w_m = w_j.$$

□

Definition 2.2 (Exceptional variation). Set

$$V_{\text{good}}(w) = \sum_{\substack{1 \leq i \leq m \\ t_i \notin \mathcal{B}_{X,J}}} |d_i|, \quad V_{\text{bad}}(w) = \sum_{\substack{1 \leq i \leq m \\ t_i \in \mathcal{B}_{X,J}}} |d_i|.$$

3 The weighted return

Put

$$\varepsilon_X = (\log X)^{-\kappa_0} + 2^{-J} + \frac{q}{T}. \quad (4)$$

Theorem 3.1 (Exceptional-variation return). *Assume the determinant-two and period envelope of TPC-159 and $T \leq X$. Then*

$$\boxed{\frac{q}{T} \left| \sum_{i=1}^m \sigma_i w_i \right| \ll \|\rho\|_\infty \left\{ \varepsilon_X V_{\text{good}}(w) + \left(1 + \frac{q}{T}\right) V_{\text{bad}}(w) \right\}.} \quad (5)$$

Proof. If $t_i \notin \mathcal{B}_{X,J}$, then it lies in the endpoint range of TPC-159 and avoids the shadow. That theorem gives

$$\frac{q}{t_i} |A_i| \ll \|\rho\|_\infty \left\{ (\log X)^{-\kappa_0} + 2^{-J} + \frac{q}{t_i} \right\}.$$

Since $t_i \leq T$, this implies

$$\frac{q}{T} |A_i| \ll \|\rho\|_\infty \varepsilon_X.$$

At a bad endpoint, use only

$$|A_i| \leq \|\rho\|_\infty \left(\frac{T}{q} + 1 \right).$$

Insert these two estimates into (3) and separate the variation according to [definition 2.2](#). □

Corollary 3.2 (Conditional logarithmic promotion). *Suppose*

$$2^J \sqrt{X} \leq T \leq X$$

and an actual source-locked registry proves, for some $\beta, \gamma > 0$,

$$V_{\text{good}}(w) \leq (\log X)^{\beta+o(1)}, \quad V_{\text{bad}}(w) \leq (\log X)^{-\gamma+o(1)}$$

and

$$\beta < \min\{\kappa_0, A\}.$$

Then the literal weighted periodic core has logarithmic exponent

$$\min\{\min(\kappa_0, A) - \beta, \gamma\} + o(1).$$

Indeed, $q \leq (\log X)^{n_0}$ and the terminal-scale lower bound make q/T smaller than every fixed negative power of $\log X$. The conclusion then follows by inserting the two variation estimates in (5). Without a terminal-scale lower bound, the factor $1 + q/T$ in the bad-variation term cannot simply be discarded.

The corollary is a promotion rule. The current occurrence archive does not supply the literal w , its variation, or the required support theorem. Its fixed- X -power exponent remains zero, even when the displayed logarithmic exponent is positive.

4 The atomic all-prefix barrier

For $1 \leq k \leq m$, consider the literal prefix cutoff

$$w_i^{(k)} = \mathbf{1}_{\{i \leq k\}}.$$

Proposition 4.1 (One-atom derivative). *The differences in (2) satisfy*

$$d_i^{(k)} = \mathbf{1}_{\{i=k\}},$$

and hence

$$V_{\text{bad}}(w^{(k)}) = \mathbf{1}_{\{t_k \in \mathcal{B}_{X,J}\}}. \quad (6)$$

Proof. The step is constant except for its single drop after index k ; when $k = m$, the terminal term $d_m = w_m$ is the same atom. $\square$

Let $\mathcal{T}_X$ be the actual registry of requested prefix endpoints. To obtain a uniform saving for all its step cutoffs from [theorem 3.1](#), one must prove

$$\boxed{\mathcal{T}_X \cap \mathcal{B}_{X,J} = \emptyset,} \quad (7)$$

or add a pointwise theorem covering the intersection. The logarithmic measure of the shadow cannot imply (7). More precisely, enlarging the source exceptional set by

$$\{2^{-J}T : T \in \mathcal{T}_X \cap [2^J\sqrt{X}, X]\}$$

does not change its continuous logarithmic measure, does not invalidate the source theorem outside the enlarged set, and places the displayed registry endpoints in the $j = J$ shadow copy. This is the atomic firewall of TPC-150 [1], now tied to the exact dyadic shadow forced by the prefix proof.

5 Phase and ledger interactions

A rational or sufficiently major-arc phase may enter the periodic factor using TPC-158 [3]; a generic phase remains outside this theorem. Alternatively, TPC-157 may charge a literal multiplier through periodic approximation [2]. These costs must be counted once. None converts a power of $\log X$ into a positive fixed- X exponent.

The accompanying exact-rational audit verifies Abel summation, the good/bad variation partition, and the one-atom prefix derivative. It records the production weight and endpoint registry as NOT-TESTABLE.

6 Conclusion

The almost-endpoint theorem can now consume a literal weight through one precise quantity: its variation on the dyadic bad set. Smooth or well-distributed weights may pass this interface; atomic prefixes expose the unresolved endpoint directly. The next advance must therefore come from actual provenance plus a variation certificate, proved shadow avoidance, or a genuinely pointwise fixed-two theorem.

References

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